Promotion Before Autonomy

Epistemic State Transitions as the Missing Primitive of Recursive AI Research Systems

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Abstract

Recursive AI research systems are often designed around generation, retrieval, graph traversal, and provenance. Those functions are necessary but insufficient. A system that can produce a new research object, follow its open questions, select a promising frontier, and preserve the resulting archive still faces a prior problem: under what conditions may an object change epistemic status? A retrieved statement can be unverified or verified; a lexical resemblance can be a candidate relation or a supported relation; an unresolved address can be malformed, ambiguous, broken, or a deliberately preserved ghost; a paper-level synthesis can become a new hypothesis without becoming evidence for itself. I call the operation that governs these changes epistemic promotion. Reading a four-layer prompt stack as a designed research artifact, this paper argues that promotion—not generation—is the missing control primitive of recursive AI research. Provenance standards describe how entities, activities, and agents participate in derivations, while content-addressed systems stabilize exact objects. Neither, by itself, specifies the legal transitions by which a research system may treat a candidate as evidence, a resemblance as a relation, or an interpretation as a new object of inquiry. A promotion layer makes these transitions explicit, typed, receipted, and reversible. The resulting architecture preserves uncertainty without immobilizing inquiry and permits autonomy without allowing inference to silently harden into evidence.

Keywords: recursive research, provenance, epistemic state, AI-assisted scholarship, research graphs, content addressing, human-AI systems

1. The hard problem begins after generation

A research agent can be made impressively busy with surprisingly little architecture. Give it sources, ask it to extract claims, let those claims spawn questions, connect the outputs in a graph, and direct the model toward whatever edge appears most promising. The system will generate. The difficult question is what the generated objects are *allowed to become*.

The prompt stack examined here separates four operations that are often collapsed under the word "research." A source-level forager reads for mechanisms, contradictions, formalism, and counterevidence. A recursive child engine takes one research object and follows its unresolved fields into new sources. A graph daemon surveys the active frontier and chooses what should be investigated next. A compiler preserves evidence, resolves relations, constructs publication views, and writes a paper. The important discovery is not that these are four prompts. It is that they are four control layers with different permissions. The reader can reinterpret a source. The child engine can extend a lineage. The daemon can allocate attention. The compiler can preserve and

AI-augmented working paper
Research and assembly record: Appendix A ·
SOURCE MAP · RETURN PATH

publish. None of those operations, however, is identical to deciding that an object has earned a new epistemic status.

That missing operation appears everywhere once one looks for it. When may an unresolved string be called a meaningful "ghost" rather than a typo? When does a shared term become a graph edge rather than a search hint? When is a generated child admitted to the research state? When can a claim move from unverified receipt to evidence? When can a paper's new synthesis return to the graph without the paper citing itself into truth? These are not edge cases. They are the gates through which an autonomous research process becomes trustworthy or becomes self-confirming.

2. Provenance answers "where from," not "what status now"

The W3C PROV data model offers a useful contrast. PROV models provenance through entities, activities, and agents; entities are used or generated by activities, and agents can bear responsibility for those activities and entities. That vocabulary is powerful because it makes derivation explicit. It can tell us that a research note was generated by an extraction activity that used a source and was associated with a model or researcher. It can represent chains of production rather than merely attaching a generic "AI-assisted" label.

But derivation is not the same as epistemic promotion. Two entities may have perfectly specified provenance while occupying very different evidentiary states. A verbatim quotation recovered from a primary source, a model's paraphrase, a speculative relation between two notes, and a paper-level inference can each have an origin trail. The provenance trail does not itself decide which may support a substantive claim. The difference is not a missing timestamp or agent. It is a missing rule of admissibility.

This distinction becomes crucial in human-AI research because the same computational system may both *produce* an object and *judge* whether that object should be trusted. If provenance is the only guardrail, generation history can be immaculate while epistemic circularity remains intact. A model can derive B from A, derive C from B, and later encounter C as if it were independent support for A unless the system carries status and lineage strongly enough to block that promotion.

The prompt stack already senses this danger. It repeatedly insists that source provenance remain distinct from inquiry lineage, that interpretations remain revisable, and that a paper is a "current wager" rather than evidence. Those rules point toward a general principle: provenance records causal history; promotion governs epistemic permission.

3. Exact identity does not solve epistemic identity

A second nearby technology is content addressing. Git's object store is content-addressable: exact stored content is retrieved through an identifier derived from that content. This solves a real archival problem. Exact duplicate payloads can be recognized mechanically rather than trusted to filenames or titles. In a portable research field, that is invaluable. It lets an archive preserve one logical evidentiary object while materializing several independently readable representations and later checking whether they drifted.

But content identity is not epistemic identity. A byte-identical claim copied from an unverified model output and from a verified primary source has the same textual content but different evidentiary history. Conversely, a corrected transcription differs at the byte level from the earlier one while remaining part of the same scholarly correction lineage. The research stack therefore needs several identities at once: exact content identity, original/local identity, appearance identity, and lineage identity. "The same note" is not a meaningful question until one specifies the transformation under which sameness is being tested.

This matters for promotion because status attaches to more than bytes. Verification may apply to a claim-as-sourced, not to every identical string wherever it appears. A promotion receipt therefore needs to identify not only *what content* was examined but *which appearance, source relation, and test* justified the transition. Content hashes can stabilize the object under inspection; they cannot supply the justification for promoting it.

4. The graph needs a quarantine zone

The graph daemon introduces another collision. To discover unexpected connections, it is instructed to infer candidate edges from shared vocabulary, related questions, competing formal shifts, recurring sources, and other weak signals. This is exactly what a useful research controller should do. A graph restricted to already-proven relations would be a poor engine for discovery.

The archival compiler, however, is intentionally conservative. It preserves literal relations, avoids fuzzy matching, distinguishes resemblance from genealogy, and forbids invented graph targets. If both instructions operate on one undifferentiated edge set, one of them must fail. Either the daemon becomes timid and stops proposing speculative relations, or the archive begins laundering heuristics into durable structure.

The solution is not to choose between exploration and rigor. It is to type the transition. A candidate edge can exist in a hypothesis graph that is visible to the scheduler but does not yet count as a supported field relation. Research can then promote, retain, or reject it. The significant move is to preserve the candidate's origin and test even when the candidate is later discarded. Negative results are part of the controller's state because they prevent the daemon from repeatedly rediscovering the same seductive false bridge.

A minimal edge-state model might distinguish LITERAL, DERIVED-DETERMINISTIC, HYPOTHESIS, SUPPORTED-INFERENCE, and REJECTED. The labels are less important than the rule: no speculative relation enters the durable field without a receipt showing what operation licensed the change. This makes it possible to exploit weak resemblance without confusing resemblance with evidence.

5. Recursive closure creates representational debt

The recursive child engine imposes a strict invariant: every child must preserve exactly the same field names and order as the parent so that every output can immediately become a new input. This is an elegant computational property. Formally, the transformation is closed over the zettel type. Operationally, it eliminates schema migration from the inner research loop.

The price is representational debt. A system whose purpose is to discover distinctions it did not know to ask for is prohibited from discovering that it needs a new field. New epistemic states must either be squeezed into existing fields or stored in a shadow structure outside the object. The later daemon and compiler already do this: they maintain active frontiers, visited edges, contradiction sets, source hubs, resolution states, derivation statuses, appearances, and reconstruction state without putting those objects into the recursive zettel schema.

This suggests a cleaner boundary. The immutable research object should remain narrow, but forageability should be a capability, not an intrinsic property of every preserved payload. Foreign or thin research objects can remain exact while an adapter provides the question, tension, missingness, and test interface needed for recursive inquiry. Likewise, promotion state can live in typed external records that point to immutable payloads. The object need not mutate for the system around it to become more expressive.

This move also resolves a deeper inconsistency between the daemon and the compiler. The daemon describes a homogeneous graph of zettels; the compiler preserves a heterogeneous field containing zettels, platforms, concepts, ghosts, sources, resources, and prompts. These need not be competing ontologies. They can be two projections: an archival field and an inquiry graph. The promotion layer mediates what moves between them and under what status.

6. Promotion is a state transition with a receipt

The recurrent pattern can now be stated more precisely. Research objects begin in a state. An operation acts on them. The operation either leaves the state unchanged or proposes a transition. A transition is legal only when its required receipt exists. Some transitions are deterministic; others require judgment; some are reversible; none should silently rewrite preserved evidence.

For example:

- RETRIEVED-CLAIM → UNVERIFIED can occur on capture.
- UNVERIFIED → VERIFIED requires a source check whose target and method are recorded.
- UNRESOLVED-ADDRESS → GHOST requires evidence that the string is an intentional intellectual address rather than malformed text; otherwise AMBIGUOUS or MALFORMED remains available.
- CANDIDATE-RELATION → SUPPORTED-RELATION requires a discriminating test or source-level support; failure may yield REJECTED while preserving the attempted edge.
- PAPER-INFERENCE → HYPOTHESIS-ZETTEL requires no new external evidence, because it is merely a change of *research role*. HYPOTHESIS-ZETTEL → EVIDENTIARY-CLAIM does require new evidence.

The last transition is the most important. A research system should be able to become new without allowing its novelty to authenticate itself. The paper can make a synthesis that no individual note contains. If that synthesis matters, it should return to the system as a hypothesis with an executable test, not as evidence. That one rule breaks a common self-confirming loop: interpretation can generate inquiry, but interpretation cannot promote itself into proof.

A promotion receipt should therefore minimally record the object, prior state, operation, resulting state, responsible agent/activity, evidence or test used, reversibility, and timestamp or checkpoint. PROV can represent much of the causal chain around that receipt; the promotion vocabulary supplies the domain-specific permission that PROV intentionally leaves open.

7. Autonomy should optimize over transitions, not just topics

The graph daemon currently asks which unresolved edge has the greatest expected epistemic gain. That is the right scheduling question, but it remains underspecified. "Epistemic gain" is a judgment unless the system can say what change would count as gain. The promotion model supplies a concrete partial answer.

A frontier edge is valuable when research on it is likely to move an important object from a weaker state to a more discriminating one: ambiguous to resolved; hypothesis to rejected or supported; unverified to verified; competing readings to a smaller surviving set. The scheduler can therefore estimate value over possible state transitions, not merely over interesting topics. This does not require reducing scholarship to a single numeric utility function. It requires the controller to name the prospective change.

That requirement has two advantages. First, it makes stopping conditions sharper. If available sources cannot discriminate the surviving states, the daemon can stop without pretending the question is resolved. Second, it makes autonomy auditable. A later reader can see not merely that the model chose to investigate a branch, but what status change it was attempting to earn and whether the attempt succeeded.

8. Operational provenance belongs beside evidence, not inside it

The prompt stack makes another important distinction when it states that its own instruction grammar is a recipe rather than a source. The prompts can determine which materials are retrieved, which contradictions are noticed, which branch is explored, and when a paper is rewritten. They are causally consequential. They are not therefore evidence for the external claims the paper makes.

This is operational provenance: a record of the decision procedures that shaped the research state without treating those procedures as substantive support. A complete publication should expose this provenance because source lists alone cannot explain why one source was retrieved while another was ignored. Yet the scholarly prose should not be forced to narrate the workflow on every page. The right architecture separates the low-bandwidth public disclosure from the detailed making history while keeping both reachable.

Operational provenance also limits what reproducibility can mean. Preserving the prompt does not preserve every latent model judgment. Two executions may rank the same frontier differently. Reproducibility therefore has levels: exact artifact reconstruction, deterministic relation reconstruction, procedural replay, and outcome replication are different claims. The promotion receipts help because they make the consequential judgments explicit even when the model's hidden deliberation is not recoverable.

9. Limits

This argument is derived from a designed prompt stack and a small associated field of research notes, not from a comparative evaluation of deployed autonomous research agents. The proposed promotion vocabulary has not yet been implemented or benchmarked. Some transitions that appear crisp on paper will require irreducible scholarly judgment. A universal scalar confidence score would be a mistake: quotation verification, graph-edge support, address resolution, and hypothesis testing demand different receipts.

The architecture also leaves open governance questions. Who may promote an object? Can a model verify a model-generated claim if it consults an independent primary source? Which promotions require human review? How should conflicting promotions from different agents coexist? The present claim is narrower: these questions become tractable only after promotion itself is represented as an operation rather than hidden inside prose or orchestration state.

Conclusion

Recursive research systems are usually described by what they can generate, retrieve, link, and remember. Those verbs describe motion through a research space; they do not specify the gates that prevent motion from becoming epistemic inflation. The prompt stack examined here already contains most of the ingredients of a rigorous system: immutable payloads, source-sensitive reading, recursive questions, a global frontier, conservative graph resolution, provenance, reversible interpretation, and explicit uncertainty. Its recurring contradictions point to one missing primitive shared across all of them.

That primitive is promotion. Promotion asks not "what comes next?" but "what has this object earned the right to become?" Once the transition is explicit, provenance can record how it happened, content identity can stabilize what changed, the graph can quarantine hypotheses without losing them, the daemon can schedule discriminating work, and the paper can generate new questions without citing itself into evidence. Autonomy then becomes less a matter of letting the system run longer and more a matter of making every consequential epistemic transition inspectable.

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Appendix A — Assembly Instrument

Exact assembly instrument: `SLIPCASE_FINAL_PROMPT.txt`. The package also preserves consequential prompt copies under `_PROMPTS/`, a claim-source map, making history, return path, bibliography, and rebuild instructions. This manuscript was assembled with AI assistance from a preserved research corpus; the assembly prompt and provenance record accompany the publication for inspection and reproduction.